

Advancing Pan-Arabic Dialect Detection on the IADD Corpus: A Comparative Study of Improved Baselines, Transformer Fine-Tuning, LLM Inference Pipelines and Hybrid Stacking Architectures

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Abstract

Arabic Dialect Detection (ADD) remains a challenging natural language processing (NLP) task due to the diglossic nature of Arabic, pervasive class imbalance in integrated datasets, and the lack of orthographic standardization in Dialectal Arabic. This paper presents a comprehensive, three-track comparative study on the Integrated Arabic Dialect Dataset (IADD; 135,804 samples) covering dialect identification at both the region level (5–6 classes) and the country level (9–10 classes). **Track 1** establishes improved deep learning baselines: simple RNN, Bidirectional LSTM (Bi-LSTM), and CNN, all augmented with pre-trained AraBERT embedding injection (768-dimensional subword vectors), achieving up to 83.70% regional accuracy (Weighted F1: 0.853, MCC: 0.719). **Track 2** fine-tunes AraBERT (`bert-base-arabertv02`; 135M parameters) for both regional (Acc: 90.94%, Macro-F1: 0.744) and country-level classification (Acc: 86.42%, Macro-F1: 0.839), and develops a QLoRA inference pipeline for the Qwen2-1.5B large language model. **Track 3** replicates the state-of-the-art hybrid stacking architecture of [Saleh et al. \(2025\)](#), combining AraBERT and CAMELBERT-DA base models with a Random Forest meta-learner, achieving 92.34% accuracy and a Weighted F1 of 0.922, thus closely approximating the 93.06% accuracy reported in the original paper, although without the application of the full cross validation loop. Across all tracks we report accuracy, precision, recall, macro and weighted F1, and Matthews Correlation Coefficient (MCC). Our findings validate that transformer-based stacking consistently outperforms single-model fine-tuning, while baselines improve with AraBERT embeddings with computational efficiency.

1 Introduction

Arabic is a highly diglossic language: Modern Standard Arabic (MSA) serves as the formal written standard taught in schools and used in media, while hundreds of millions of speakers communicate daily in geographically and culturally distinct spoken dialects collectively termed Dialectal Arabic (DA). In the digital sphere (social media, web forums, messaging platforms, ...), users overwhelmingly write in DA ([Abu Kwaik et al., 2018](#)). This divergence from MSA in vocabulary, morphology, syntax, and loanword inventory poses fundamental challenges for NLP systems trained primarily on MSA corpora.

Accurate Arabic Dialect Detection (ADD) is a critical upstream component for downstream tasks including sentiment analysis, machine translation, hate-speech detection, and question answering tailored to specific Arabic-speaking communities. The task is inherently difficult for several reasons: (i) dialects of neighboring countries (e.g., Palestinian and Jordanian within the Levantine region) share substantial vocabulary; (ii) code-switching between dialects and MSA is common; (iii) Arabic’s agglutinative morphology produces extreme vocabulary diversity; and (iv) naturally-collected corpora exhibit severe class imbalance.

This project conducts a systematic multi-track evaluation on the **Integrated Arabic Dialect Dataset (IADD)** ([Zahir, 2022](#)), covering a progression from classical deep learning architectures through modern transformer fine-tuning and culminating in a replication of a hybrid stacking approach from recent literature ([Saleh et al., 2025](#)). The three tracks are:

- **Track 1 (Tarek):** Improved deep learning baselines (Simple RNN, Bi-LSTM, CNN) with pre-trained AraBERT embedding injection via the HuggingFace `aubmindlab/bert-base-arabertv02` model.
- **Track 2 (Mohamad):** Fine-tuned AraBERT (encoder-only) for region- and country-level classification, plus a QLoRA inference pipeline for the Qwen2-1.5B LLM (similar to Llama architecture - decoder only).
- **Track 3 (Ayman):** Replication of Saleh et al. (2025)’s hybrid stacking of AraBERT and CAMELBERT-DA with a Random Forest meta-learner.

All code and notebooks are publicly available at <https://github.com/DoubleOminou/csc464-final-project>.

2 Dataset and Exploratory Analysis

2.1 The IADD Corpus

The Integrated Arabic Dialect Dataset (Zahir, 2022) harmonizes five constituent corpora: DART (social media), SHAMI (Levantine), PADIC (North African), AOC (Online Commentary), and TSAC (Tunisian), yielding **135,804 raw samples**. Each record carries four fields: **Sentence** (raw Arabic text), **Region** (6-class coarse label), **Country** (9-class fine label), and **DataSource**.

(Full results in Table 1, Appendix 9.)

The corpus exhibits severe class imbalance: Levantine alone accounts for 64.5% of samples, while Iraqi constitutes only 0.2% (216 samples). At the country level (9 classes after dropping 16,441 “NA” Gulf/General entries), Syria dominates with 44,972 samples while Iraq provides only 216, constituting a 208:1 ratio.

2.2 Text Statistics

(Full results in Table 2, Appendix 9.)

Egyptian texts are notably longer (mean 24 words) and more variable, while Maghrebi texts are shortest (mean 7.6 words). The Gulf region exhibits extreme standard deviation in character count (756.69) due to a small number of very long documents.

3 Preprocessing and Tokenization

Tracks 1 and 2 remove empty rows and rely on the AraBERT tokenizer’s internal normalization, whereas Track 3 employs an explicit and detailed pipeline: raw text is cleaned by removing URLs, user mentions (@username), HTML entities, non-Arabic characters, and repeating punctuation; Arabic-specific normalization is applied via the `pyarabic` library: diacritics (tashkeel) are stripped, alef variants are normalized to a bare alef, and trailing taa marbuta is normalized to ha.

Subword tokenization: A key architectural decision across all tracks is the use of the **AraBERT tokenizer** (`aubmindlab/bert-base-arabertv02`) with a vocabulary of 64,000 WordPiece subword units, rather than standard word-level tokenization. Arabic’s agglutinative morphology, where prefixes, clitics, and suffixes attach to roots to form complex surface forms, leads to extreme vocabulary explosion under word-level tokenization and excessive OOV rates. Subword tokenization decomposes Arabic morphemes consistently, resulting in a vocabulary coverage exceeding 95% across all dialectal regions.

After preprocessing in Track 3, 6,582 rows (4.8%) became empty strings and were discarded, leaving **129,222 usable samples**. Tracks 1 and 2 retain all six region classes (including Iraqi), whereas Track 3 differs in its treatment of the Iraqi class (see Section 4).

4 Experimental Setup

4.1 Data Splits

(Full results in Table 3, Appendix 9.)

Tracks 1 and 2 use a 70/15/15 train/validation/test stratified split and retain the Iraqi class (151 training samples). Track 3 follows Saleh et al. (2025) by excluding Iraqi (below 1% threshold) and uses a 75/25 split without a separate validation fold, relying on early stopping and best-epoch checkpointing.

4.2 Evaluation Metrics

All models are evaluated with six metrics: **Accuracy**, **Weighted Precision**, **Weighted Recall**, **Weighted F1** (reflecting dataset-level performance), **Macro F1** (unweighted

average across classes, sensitive to minority class performance), and **Matthews Correlation Coefficient (MCC)** (robust to class imbalance; ranges from -1 to $+1$). Given the 64.5% Levantine dominance, Macro F1 and MCC are the primary indicators of a model’s true dialectal discrimination capability.

4.3 Compute Environment

All models were trained on NVIDIA Tesla T4 GPUs (15.6 GB VRAM) via Google Colab (Tracks 1–2) and Kaggle (Track 3). Framework versions: PyTorch 2.10.0+cu128; HuggingFace Transformers; Keras 3 (TensorFlow backend for Track 1 baselines).

5 Track 1: Improved Deep Learning Baselines

5.1 Architecture and Design Choices

Track 1 establishes robust baselines using three classical sequence architectures: **Simple RNN**, **Bidirectional LSTM (Bi-LSTM)**, and **CNN with 1-D convolutions**. The critical innovation is replacing random embedding initialization with **pre-trained AraBERT embedding injection**:

1. The full aubmindlab /bert-base-arabertv02 model (135M parameters) is loaded via HuggingFace.
2. The 768-dimensional word embedding matrix (shape $64,000 \times 768$) is extracted from `model.embeddings.word_embeddings.weight`.
3. This matrix is injected into the Keras Embedding layer of each baseline model with `trainable=False` (frozen) during initial epochs.

Text is tokenized using the AraBERT tokenizer and truncated/padded to `MAX_LEN_SEQ = 100` tokens. Batch size is 64, and training runs for up to 15 epochs with early stopping on validation loss (patience=3). MSA stop-words are not removed in this track (the models leverage full subword context). Balanced class weights are applied via `sklearn.utils.class_weight.compute_class_weight` to address imbalance.

Architecture specifics:

- **Simple RNN:** Single RNN layer (64

units, 0.2 recurrent dropout) $\rightarrow$ dropout (0.5) $\rightarrow$ dense(64) ReLU $\rightarrow$ dense softmax.

- **Bi-LSTM:** Bidirectional LSTM (64 units; 32 units) $\rightarrow$ dropout (0.5) $\rightarrow$ dense(64) ReLU $\rightarrow$ dense softmax.
- **CNN:** 1-D convolution branch (filter: 128; kernel size: 5) $\rightarrow$ global max pooling $\rightarrow$ dropout (0.5) $\rightarrow$ dense(64) ReLU $\rightarrow$ dense softmax.

5.2 Regional Classification Results (6 Classes)

(Full results in Table 4, Appendix 9.)

(Full results in Table 5, Appendix 9.)

The Simple RNN completely fails on Gulf ($F1=0.00$)—the model degenerates into majority-class prediction for many classes. The Bi-LSTM recovers substantially through bidirectional context, and the CNN achieves the highest overall performance through n-gram feature extraction. Notably, the CNN achieves the best Iraqi F1 (0.51) despite only 32 test samples, suggesting local pattern features are more discriminative for this minority class than sequential modeling.

5.3 Country Classification Results (10 Classes)

(Full results in Table 6, Appendix 9.)

Interestingly, the Bi-LSTM outperforms CNN at the country level (78.1% vs. 76.7% accuracy), suggesting that sequential modeling better captures the finer within-region linguistic distinctions (e.g., Palestinian vs. Syrian vs. Lebanese). Tunisia achieves the highest country-level F1 across all baseline models (CNN: 0.91, Bi-LSTM: 0.92), attributable to the distinctive TSAC sub-corpus features.

6 Track 2: Transformer Fine-Tuning and LLM Pipeline

6.1 AraBERT Fine-Tuning

Model: aubmindlab/bert-base-arabertv02 — a BERT-large-uncased architecture pre-trained on 77GB of Arabic text (news, Wikipedia, social media) using a 64,000-token WordPiece vocabulary with Arabic-specific normalization. Total parameters: **135,197,958** (fully fine-tuned), slight shift

between region and country models. A linear classification head ($768 \rightarrow k$ classes) replaces the MLM head.

Hyperparameters: 3 epochs; batch size 32; learning rate 2×10^{-5} with 10% linear warmup (891 steps for region, 783 for country); weight decay 0.01; AdamW optimizer. Best checkpoint loaded via weighted F1 on validation set. Balanced class weights are computed and passed to a custom cross-entropy loss.

6.1.1 Regional Classification (6 Classes)

(Full results in Table 9, Appendix 9.)

AraBERT provides a dramatic improvement over the best baseline (CNN: 83.7% $\rightarrow$ AraBERT: 90.9%), with Macro F1 jumping from 0.610 to 0.744. Notably, AraBERT achieves 0.77 F1 on the Iraqi class (32 test samples), outperforming the CNN baseline (0.51), demonstrating the power of pre-trained contextual representations for extreme minority classes. The “General/MSA” class remains the hardest (F1=0.31), as it lacks consistent dialectal markers.

6.1.2 Country-Level Classification (9 Classes)

(Full results in Table 10, Appendix 9.)

The most confused country pairs are Jordan/Palestine (both Levantine) and Jordan/Lebanon. Tunisia achieves the highest country-level F1 (0.95) due to the linguistically distinctive TSAC corpus. Iraq, despite extreme sparsity (32 test samples), achieves F1=0.79 under AraBERT’s balanced weighting.

(Full results in Table 11, Appendix 9.)

Interestingly, country-level Macro F1 (0.839) exceeds region-level Macro F1 (0.744), because distinguishing 9 country classes compels the model to learn richer dialect-specific features, and balanced weighting forces better minority-class performance.

6.2 Qwen2-1.5B QLoRA Pipeline

As an exploratory Track 2 component, a complete fine-tuning and inference pipeline was developed for the **Qwen2-1.5B** open LLM (1,548.1M total parameters; 151,643-token vocabulary). Qwen2 was selected over Llama due to its superior multilingual capability and

native Arabic comprehension.

QLoRA configuration: 4-bit NF4 quantization via BitsAndBytes; double quantization enabled; compute dtype bfloat16. LoRA adapters (rank $r=16$, $\alpha=32$, dropout 0.05) applied to Q, K, V, and O projection matrices of all attention layers. **Trainable parameters:** **4,367,360 / 1,548,090,880 = 0.28%** of total, enabling efficient fine-tuning on 15.6 GB VRAM. Training uses gradient accumulation of 4 steps (effective batch size 32), LR= 2×10^{-4} , and 5% warmup (1,782 steps). Hardware constraints (T4 VRAM) precluded full test-set evaluation within the runtime; the pipeline is validated for correctness and serves as the foundation for future large-scale evaluation.

7 Track 3: Hybrid Stacking Transformer Architecture

7.1 Overview

Track 3 replicates the state-of-the-art methodology of Saleh et al. (2025). The core idea is that a single transformer exhibits representational blind spots; AraBERT (monolingual) excels at MSA-proximate dialectal features while a multilingual/dialectal model captures cross-lingual topological features. Stacking their outputs into a meta-learner harnesses complementary strengths.

7.2 Data Preparation

Following Saleh et al. (2025), the Iraqi class is excluded (216 samples, <1%), leaving **129,006 samples across 5 classes** (EGY, GLF, LEV, MGH, general). Preprocessing uses the full Arabic pipeline from Section 3 plus MSA stop-word removal. A 75/25 stratified split yields 96,754 training and 32,252 test samples.

Train class distribution: LEV 65,646 (67.8%) $\gg$ MGH 20,595 (21.3%) $>$ GLF 5,012 (5.2%) $>$ EGY 3,626 (3.7%) $>$ general 1,875 (1.9%).

7.3 Level-1 Base Models

Both base models are fine-tuned using the HuggingFace Trainer with identical hyperparameters (except model identity): up to 5 epochs (early stopping, patience=2), batch

size 16, LR= 2×10^{-5} , AdamW with weight decay 0.01, balanced class weights, and patience-based best-checkpoint selection (accuracy).

Word2Vec CBOW embeddings (for DL baselines): A 128-dimensional CBOW Word2Vec model is trained on the 96,754 training sentences (vocabulary: 58,931 unique tokens), providing baseline DL embeddings.

(Full results in Table 13, Appendix 9.)

Note: Due to an unavailability issue with the originally planned XLM-R-Dialectal model (rabiaqayyum/xlm-roberta-base-finetuned-arabic-dialect), the CAMELBERT-DA model was used as a drop-in replacement, a BERT-base model pre-trained specifically on dialectal Arabic, providing similar cross-dialect generalization.

7.3.1 Level-1 Fine-Tuning Results

(Full results in Table 14, Appendix 9.)

7.3.2 Level-1 Per-Class F1

(Full results in Table 15, Appendix 9.)

CAMELBERT-DA slightly outperforms AraBERT on Gulf (0.77 vs. 0.75) and general (0.26 vs. 0.18), suggesting that dialectal pre-training helps with minority and cross-dialect classes.

7.4 Level-2 Meta-Learner: Random Forest Stacking

The probability distributions (softmax logits, shape $N \times 5$) from both Level-1 models are concatenated into a $N \times 10$ stacking feature vector per sample. This is used to train and evaluate a **Random Forest** meta-classifier:

- 300 estimators; unconstrained depth; `class_weight="balanced"`; all CPU cores.
- Input features: [AraBERT_{EGY}, AraBERT_{GLF}, AraBERT_{LEV}, AraBERT_{MGH}, AraBERT_{Gen}, XLM-R_{EGY}, XLM-R_{GLF}, XLM-R_{LEV}, XLM-R_{MGH}, XLM-R_{Gen}].
- Feature importance analysis (Section 8) reveals model contributions.

7.5 Stacking Results

(Full results in Table 16, Appendix 9.)

(Full results in Table 17, Appendix 9.)

7.5.1 Epoch-by-Epoch Training Dynamics

(Full results in Table 18, Appendix 9.)

Both LSTM and GRU peak at epoch 2 on the test set (90.67% and 90.52% respectively) before overfitting. The training accuracy continues to rise to >98% by epoch 10, indicating memorization of the training distribution without further generalization, a known limitation of recurrent models on noisy social-media text.

7.6 Paper Comparison: Replication vs. Saleh et al. (2025)

(Full results in Table 19, Appendix 9.)

Our replication surpasses the paper’s baseline models across the board, likely due to higher-quality Word2Vec CBOW embeddings (128-dim vs. the paper’s simpler embeddings) and aggressive balanced class weighting. The stacking result falls 0.72% short of the paper’s reported 93.06%, which we could attribute to the lack of a full 5-fold cross validation run on the final stack due to runtime limitations. Implementing the full run (not just single split) constitutes a significant next step for future work, as it may mitigate the potential data leakage caused by using train softmax probabilities directly to fit the RF meta-learner.

8 Discussion

8.1 Feature Importance in the RF Meta-Learner

The Random Forest meta-learner reveals which model’s predictions are most informative for dialect discrimination:

(Full results in Table 22, Appendix 9.)

CAMELBERT-DA’s probability outputs dominate (top 4 features, combined importance 0.598), indicating that dialectal pre-training provides stronger discriminative signal for minority classes (EGY, GLF) than the general-purpose AraBERT. The EGY and GLF features carry the highest importance, consistent with their status as minority classes where confident discrimination matters most.

8.2 Minority Class Analysis

The “general/MSA” class is consistently the most difficult across all models:

(Full results in Table 23, Appendix 9.)

The progression from 0.03 (CNN) to 0.31 (Stacking) illustrates the compound benefit of stacking: the RF meta-learner learns to combine both transformers’ uncertainty signals for ambiguous “general” inputs that mix MSA with regional features.

8.3 Cross-Track Performance Summary

(Full results in Table 24, Appendix 9.)

8.4 Key Findings

Finding 1 — Embedding injection is highly effective: Injecting 768-dimensional AraBERT embeddings into lightweight baselines (CNN, Bi-LSTM) raises their regional accuracy to 79–84%, a substantial improvement over typical random-initialization baselines (~60–70%).

Finding 2 — Self-attention captures long-range dialect signals: Full AraBERT fine-tuning (+6.5% over CNN baseline) confirms that bidirectional context modeling better isolates dialect-specific syntactic constructions than local n-gram or sequential features alone.

Finding 3 — Stacking provides consistent minority-class improvement: The RF meta-learner’s greatest gain over individual transformers is on the “general” class (F1: 0.26 → 0.31) and Gulf (F1: 0.75 → 0.77), the two most confused classes.

Finding 4 — Country-level Macro F1 exceeds region-level: AraBERT’s country-level Macro F1 (0.839) surpasses region-level (0.744) under balanced weighting. This is counter-intuitive but is explained by the country-level task forcing sharper feature specialization per country corpus.

Finding 5 — Dialectal pre-training dominates feature importance: CAMELBERT-DA’s outputs account for 59.8% of the RF meta-learner’s total feature importance, confirming that domain-matched pre-training (dialectal Arabic) yields more informative probability distributions than general Arabic pre-training.

9 Conclusion and Future Work

We have presented a comprehensive multi-track study of Arabic Dialect Detection on the IADD corpus, systematically comparing classical deep learning baselines (with AraBERT embedding injection), fine-tuned transformers, an LLM inference pipeline, and a hybrid stacking architecture. Our results confirm that: (1) AraBERT embedding injection provides a cost-effective 14–25% accuracy improvement over randomly initialized baselines; (2) full fine-tuning of AraBERT achieves 90.94% regional and 86.42% country-level accuracy; and (3) hybrid stacking of AraBERT with CAMELBERT-DA achieves 92.34% accuracy and 0.922 Weighted F1, closely approaching the 93.06% state-of-the-art of Saleh et al. (2025).

Future work will focus on: (i) executing the Qwen2 QLoRA pipeline on HPC clusters to benchmark zero-shot and few-shot dialect detection against fine-tuned models; (ii) applying data augmentation (back-translation, DA paraphrasing) to rescue the extreme minority classes (Iraqi: 216 samples, Gulf: 6,682); (iii) running the full cross fold process on the stacking architecture, as well as attempting a country split on it; and (iv) investigating dialectal code-switching, that is mixed MSA/DA sentences, as a more realistic evaluation scenario.

References

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Appendix A: Dataset and Experimental Setup Tables

A.1 IADD Corpus Distribution

A.2 Text Length Statistics

A.3 Dataset Splits

Appendix B: Track 1 Deep Learning Baseline Tables

B.1 Regional Classification Results

B.2 Country Classification Results

B.3 Track 1 Bi-LSTM Country-Level Detailed Classification Report

B.4 Track 1 CNN Country-Level Detailed Classification Report

Appendix C: Track 2 Transformer Fine-Tuning Tables

C.1 AraBERT Regional Classification

C.2 AraBERT Country-Level Classification

C.3 AraBERT Region vs. Country Comparison

C.4 Qwen2-1.5B QLoRA Configuration

Appendix D: Track 3 Stacking Architecture Tables

D.1 Level-1 Model Architectures

D.2 Level-1 Fine-Tuning Results

D.3 Stacking and Per-Class Results

D.4 Paper Comparison: Replication vs. Saleh et al. (2025)

D.5 Track 3 DL Baseline Training Configuration

D.6 Inference Demo Examples (Track 3 Stacking Pipeline)

The pipeline achieves high-confidence predictions for Levantine, Maghrebi, and Gulf texts. The Egyptian example shows some Levantine confusion (19.3%), reflecting shared vocabulary between Egyptian and Palestinian Arabic. The ambiguous “general/formal” sentence is misclassified as Gulf, consistent with the model’s difficulty with the general class throughout all experiments.

Appendix E: Discussion Tables

E.1 Random Forest Feature Importances

E.2 general/MSA Class F1 Progression

E.3 Cross-Track Performance Summary

Table 1: IADD class distribution and data source breakdown (full dataset, $N = 135,804$).

Region	Countries	Count	%	AOC	DART	PADIC	SHAMI	TSAC
Levantine (LEV)	Syria, Palestine, Lebanon, Jordan	87,573	64.5	6,503	397	14,426	66,247	0
Maghrebi (MGH)	Algeria, Tunisia, Morocco	33,996	25.0	28	331	21,639	0	11,998
Gulf (GLF)	Saudi Arabia et al.	6,682	4.9	6,259	423	0	0	0
Egyptian (EGY)	Egypt	4,837	3.6	4,414	423	0	0	0
General	MSA / Mixed	2,500	1.8	2,500	0	0	0	0
Iraqi (IRQ)	Iraq	216	0.2	9	207	0	0	0
Total		135,804	100					

Table 2: Text length statistics per region (word count and character count).

Region	Word Count			Character Count		
	Mean	Median	Std	Mean	Median	Std
EGY	23.98	13.0	40.40	130.75	67.0	224.37
GLF	15.72	10.0	93.11	89.24	52.0	756.69
IRQ	12.68	12.0	7.07	67.39	63.0	38.17
LEV	13.01	9.0	15.09	66.44	47.0	78.18
MGH	7.62	5.0	22.98	39.95	25.0	132.60
General	9.00	6.0	10.53	49.95	32.0	59.01

Table 3: Dataset splits across tracks. All splits are stratified by class label.

Track	Task	Classes	Train	Val	Test	Split
Track 1 (Tarek)	Region	6 (incl. IRQ)	95,062	20,371	20,371	70/15/15
Track 1 (Tarek)	Country	10 (incl. Gulf=NA)	95,062	20,371	20,371	70/15/15
Track 2 (Mohamad)	Region	6 (incl. IRQ)	95,062	20,371	20,371	70/15/15
Track 2 (Mohamad)	Country	9 (excl. NA)	83,554	17,904	17,905	70/15/15
Track 3 (Ayman)	Region	5 (excl. IRQ)	96,754	—	32,252	75/25

Table 4: Track 1 regional classification results (6 classes: EGY, GLF, IRQ, LEV, MGH, General). Test set = 20,371 samples.

Model	Accuracy	Weighted F1	MCC	Macro F1
Simple RNN	0.6243	0.6831	0.4690	0.3316
Bi-LSTM	0.7901	0.8264	0.6700	0.5540
CNN	0.8370	0.8529	0.7193	0.6100

Table 5: Track 1 regional per-class F1 scores (test set, 6-class). Bolded values are highest per column.

Model	EGY	GLF	IRQ	LEV	MGH	General
Simple RNN	0.24	0.00	0.10	0.74	0.78	0.10
Bi-LSTM	0.63	0.61	0.19	0.86	0.86	0.16
CNN	0.57	0.57	0.51	0.90	0.88	0.21

Table 6: Track 1 country classification results (10 classes including Gulf=NA). Test set = 20,371 samples.

Model	Accuracy	Weighted F1	MCC	Macro F1
Simple RNN	0.4554	0.4617	0.3819	0.3860
Bi-LSTM	0.7781	0.7821	0.7369	0.7258
CNN	0.7672	0.7699	0.7226	0.7129

Table 7: Full classification report: Track 1 Bi-LSTM Country (10-class). Test set = 20,371 samples.

Country	Precision	Recall	F1	Support
Algeria	0.83	0.87	0.85	2,164
Egypt	0.61	0.75	0.67	725
Iraq	0.43	0.62	0.51	32
Jordan	0.53	0.63	0.58	1,053
Lebanon	0.62	0.80	0.70	1,625
Morocco	0.73	0.85	0.78	1,082
Gulf (NA)	0.79	0.71	0.75	2,466
Palestine	0.65	0.67	0.66	2,679
Syria	0.92	0.79	0.85	6,746
Tunisia	0.92	0.92	0.92	1,799
Macro Avg	0.70	0.76	0.73	20,371
Weighted Avg	0.79	0.78	0.78	20,371

Table 8: Full classification report: Track 1 CNN Country (10-class). Test set = 20,371 samples.

Country	Precision	Recall	F1	Support
Algeria	0.79	0.89	0.84	2,164
Egypt	0.70	0.63	0.67	725
Iraq	0.52	0.50	0.51	32
Jordan	0.56	0.55	0.55	1,053
Lebanon	0.61	0.74	0.67	1,625
Morocco	0.80	0.75	0.77	1,082
Gulf (NA)	0.72	0.78	0.75	2,466
Palestine	0.61	0.67	0.63	2,679
Syria	0.91	0.79	0.84	6,746
Tunisia	0.91	0.90	0.91	1,799
Macro Avg	0.71	0.72	0.71	20,371
Weighted Avg	0.78	0.77	0.77	20,371

Table 9: AraBERT regional results (6-class). Test set: 20,371 samples.

Metric	Value	Class	Precision	Recall	F1
Accuracy	0.9094	EGY	0.72	0.79	0.75
Weighted F1	0.9156	GLF	0.71	0.79	0.75
Macro F1	0.7437	IRQ	0.78	0.76	0.77
		LEV	0.97	0.92	0.95
		MGH	0.92	0.96	0.94
		General	0.23	0.45	0.31

Table 10: AraBERT country-level results (9-class). Test set: 17,905 samples.

Metric	Value	Country	Precision	Recall	F1
Accuracy	0.8642	Algeria	0.89	0.90	0.89
Weighted F1	0.8658	Egypt	0.86	0.88	0.87
Macro F1	0.8388	Iraq	0.72	0.88	0.79
		Jordan	0.68	0.72	0.70
		Lebanon	0.83	0.81	0.82
		Morocco	0.79	0.88	0.83
		Palestine	0.74	0.82	0.78
		Syria	0.94	0.88	0.91
		Tunisia	0.95	0.96	0.95

Table 11: AraBERT performance comparison: Region (6-class) vs. Country (9-class) tasks.

Task	Accuracy	Weighted F1	Macro F1
Region (6 classes)	0.9094	0.9156	0.7437
Country (9 classes)	0.8642	0.8658	0.8388

Table 12: Qwen2-1.5B QLoRA hyperparameter configuration.

Parameter	Value
Base model	Qwen/Qwen2-1.5B
Total parameters	1,548,090,880
Quantization	4-bit NF4 (BitsAndBytes)
Double quantization	Enabled
Compute dtype	bfloat16
LoRA rank (r)	16
LoRA α	32
LoRA dropout	0.05
LoRA target modules	q_proj, k_proj, v_proj, o_proj
Trainable parameters	4,367,360 (0.28%)
Per-device batch size	8
Gradient accumulation	4 (effective batch: 32)
Learning rate	2×10^{-4}
Warmup steps	1,782 (5% of training)
Epochs	3
Vocab size	151,643
Max sequence length	128 tokens

Table 13: Level-1 model architectures (Track 3 stacking).

Model	HuggingFace ID	Type	Parameters
AraBERT	aubmindlab/bert-base-arabertv02	Monolingual Arabic	135M
CAMeLBERT-DA	CAMeL-Lab/bert-base-arabic-camelbert-da	Dialectal Arabic	135M

Table 14: Level-1 transformer results (Track 3, 5-class, test set $N = 32,252$).

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
AraBERT	0.9202	0.9129	0.9202	0.9135
CAMeLBERT-DA	0.9231	0.9177	0.9231	0.9189

Table 15: Level-1 per-class F1 scores (Track 3, 5-class).

Model	EGY	GLF	LEV	MGH	general
AraBERT	0.75	0.75	0.95	0.93	0.18
CAMeLBERT-DA	0.75	0.77	0.95	0.93	0.26
<i>Support:</i> EGY=1,209; GLF=1,670; LEV=21,883; MGH=6,865; general=625					

Table 16: Track 3 complete model comparison (5-class, $N = 32,252$ test samples). Highlighted row = best per column.

Model	Accuracy	Precision	Recall	Weighted F1
CNN (Word2Vec CBOW)	0.9025	0.8883	0.9025	0.8924
GRU (Word2Vec CBOW)	0.9052	0.8962	0.9052	0.8970
LSTM (Word2Vec CBOW)	0.9067	0.8971	0.9067	0.8980
AraBERT (fine-tuned)	0.9202	0.9129	0.9202	0.9135
CAMeLBERT-DA (fine-tuned)	0.9231	0.9177	0.9231	0.9189
Stacking (AraBERT + CAMeLBERT-DA + RF)	0.9234	0.9213	0.9234	0.9222

Table 17: Track 3 per-class F1 scores across all model tiers. Macro F1 (unweighted) shown in last column.

Model	EGY	GLF	LEV	MGH	general	Macro F1
CNN	0.65	0.65	0.94	0.92	0.03	0.638
GRU	0.70	0.66	0.94	0.92	0.05	0.654
LSTM	0.71	0.67	0.94	0.92	0.05	0.658
AraBERT	0.75	0.75	0.95	0.93	0.18	0.712
CAMeLBERT-DA	0.75	0.77	0.95	0.93	0.26	0.732
Stacking	0.76	0.77	0.96	0.94	0.31	0.748

Table 18: LSTM and GRU training curves (5-class, Track 3). Best test accuracy highlighted.

Epoch	LSTM			GRU		
	Train Acc	Train Loss	Test Acc	Train Acc	Train Loss	Test Acc
1	86.49%	0.3967	89.54%	87.07%	0.3770	90.11%
2	92.53%	0.2192	90.67%	92.96%	0.2042	90.52%
3	95.14%	0.1353	89.89%	95.26%	0.1341	89.71%
5	97.58%	0.0657	89.41%	97.62%	0.0632	89.39%
10	98.60%	0.0349	89.03%	98.52%	0.0359	88.93%

Table 19: Accuracy and Weighted F1 comparison: our replication vs. Saleh et al. (2025). Differences shown in the Δ columns.

Model	Saleh et al. (2025)		Ours		Δ	
	Acc	F1	Acc	F1	Δ Acc	Δ F1
CNN	87.38%	87.49%	90.25%	89.24%	+2.87%	+1.75%
GRU	88.00%	87.90%	90.52%	89.70%	+2.52%	+1.80%
LSTM	88.00%	87.90%	90.67%	89.80%	+2.67%	+1.90%
AraBERT	90.50%	90.20%	92.02%	91.35%	+1.52%	+1.15%
XLM-R/CAMeLBER-DA	91.43%	91.49%	92.31%	91.89%	+0.88%	+0.40%
Stacking	93.06%	93.18%	92.34%	92.22%	-0.72%	-0.96%

Table 20: Track 3 DL model hyperparameters (Word2Vec CBOW + PyTorch).

Parameter	Value
Embedding type	Word2Vec CBOW
Embedding dimension	128
Vocabulary size	58,931 (trained on IADD train set)
Max sequence length	100 tokens
LSTM/GRU hidden size	200 units
CNN filters	128 per kernel size
CNN kernel size	4
Dropout	0.2
Optimizer	Adam, LR= 10^{-3}
Batch size	64
Epochs	10
Loss	Cross-entropy (balanced class weights: RF)

Table 21: Sample inference outputs from the Track 3 stacking pipeline on held-out Arabic sentences.

Input (Arabic)	Predicted	Confidence
وش فيك تسألني إذا كنت غالي ويكثر جنوني	GLF	GLF: 99.7%
انت بذك تحكي معي هلق مش وقتها	LEV	LEV: 100.0%
انت فبن يا حبيبي الليلة دي	EGY	EGY: 80.3%, LEV: 19.3%
كيفاش تقدر تقول هاد الكلام بالصبح	MGH	MGH: 100.0%
هل الحكومة تستاهل الثقة الجديدة	GLF	GLF: 65.3%, LEV: 16.3%

Table 22: Top-10 Random Forest feature importances for the stacking meta-learner.

Rank	Feature	Importance
1	CAMeLBER-DA $P(EGY)$	0.1832
2	CAMeLBER-DA $P(GLF)$	0.1656
3	CAMeLBER-DA $P(MGH)$	0.1323
4	CAMeLBER-DA $P(LEV)$	0.1168
5	AraBERT $P(EGY)$	0.0983
6	CAMeLBER-DA $P(Gen)$	0.0976
7	AraBERT $P(MGH)$	0.0820
8	AraBERT $P(GLF)$	0.0491
9	AraBERT $P(LEV)$	0.0443
10	AraBERT $P(Gen)$	0.0308

Table 23: “general/MSA” class F1 progression across model tiers (Track 3).

Model	general F1
CNN (Word2Vec)	0.03
LSTM (Word2Vec)	0.05
AraBERT	0.18
CAMeLBERT-DA	0.26
Stacking	0.31

Table 24: Full cross-track summary of all models on their respective test sets. MCC reported where computed.

Track	Task	Model	Classes	Acc	W-F1	Macro-F1	MCC
T1	Region	Simple RNN (AraBERT emb.)	6	0.624	0.683	0.332	0.469
T1	Region	Bi-LSTM (AraBERT emb.)	6	0.790	0.826	0.554	0.670
T1	Region	CNN (AraBERT emb.)	6	0.837	0.853	0.610	0.719
T1	Country	Simple RNN (AraBERT emb.)	10	0.455	0.462	0.386	0.382
T1	Country	Bi-LSTM (AraBERT emb.)	10	0.778	0.782	0.726	0.737
T1	Country	CNN (AraBERT emb.)	10	0.767	0.770	0.713	0.723
T2	Region	AraBERT (fine-tuned)	6	0.909	0.916	0.744	—
T2	Country	AraBERT (fine-tuned)	9	0.864	0.866	0.839	—
T3	Region	CNN (Word2Vec CBOW)	5	0.902	0.892	0.638	—
T3	Region	GRU (Word2Vec CBOW)	5	0.905	0.897	0.654	—
T3	Region	LSTM (Word2Vec CBOW)	5	0.907	0.898	0.658	—
T3	Region	AraBERT (fine-tuned)	5	0.920	0.914	0.712	—
T3	Region	CAMeLBERT-DA (fine-tuned)	5	0.923	0.919	0.732	—
T3	Region	Stacking (RF)	5	0.923	0.922	0.748	—